

Asher Siddiqui

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EDUCATION

FAST - NUCES

Bachelor of Science in Software Engineering

Relevant Courses: Applied DevOps, Cloud Security, Agile Software Project Management

Awards & Honors: Dean's List of Honors (3.5+ SGPA) in Fall 2023, Spring 2024, Fall 2024

Karachi, Pakistan

Expected Graduation: June 2025

TECHNICAL SKILLS

Languages: Go, Python, Java, C, JavaScript

Backend & Frameworks: Node.js, Express.js, Spring Boot

Databases: MySQL, Oracle RDBMS, MongoDB, Redis

DevOps & Cloud: Docker, Kubernetes, AWS, Linux, Bash, Ansible, Prometheus, Node Exporter, Falco

EXPERIENCE

Grayhat Private Limited

Software Engineering Intern

July 2024 – Sept 2024

Islamabad (Remote)

- Developed and managed the MVP for v0.dev's VS Code extension, coordinating with one other developer to accelerate delivery and increase user satisfaction.
- Built an automated developer productivity integration system, synchronizing events across GitHub, Kimai, Activity Watch, and Plane, enabling streamlined workflow tracking and increased developer efficiency.

FEATURED PROJECTS

FYP - DDoS Detection in Kubernetes using ML and DL | *Kubernetes, Ansible, Prometheus, Falco, Go, Python* | [Github](#)

- Deployed a 3-node botnet and a 4-node bare-metal Kubernetes cluster using Ansible to simulate real DDoS attack scenarios for training ML and DL models.
- Developed a custom data collection pipeline to log physical resource-related system calls and usage metrics, integrating Falco and Node Exporter, enabling seamless training data generation.
- Developed and deployed attack simulation scripts using Python on the C&C server and the botnet infrastructure, enabling orchestration of controlled DDoS attacks.

Rate Limiter | *Go, Docker, Redis* | [Github](#)

- Engineered a Layer 4 rate limiter supporting Token Bucket and Sliding Window algorithms using Go, preventing API misuse through uncontrolled requests.
- Optimized HTTP responses by adding custom rate-limiting headers, improving client-side request handling and ensuring request quota transparency.
- Improved system scalability and response by leveraging Redis for state management, ensuring low-latency request processing under high traffic.

Load Balancer | *Go, Express.js, Docker, Git, HTML* | [Github](#)

- Designed and implemented a high-throughput HTTP reverse proxy implementing Round Robin and Least Connection algorithms using Go, achieving 2000 requests per second with minimal latency.
- Increased reliability by integrating health checks, ensuring automatic rerouting of traffic to available servers.
- Containerized the load balancer using Docker, enabling seamless deployment and scalability, and consistent performance across environments.

ADDITIONAL PROJECTS

CUDA-based Rendering of Mandelbrot Set | [Github](#)

Developed a Mandelbrot Set rendering script using CUDA-based GPU processing, achieving 82% speed improvement over a CPU-based approach.

Project Management System

Engineered a secure three-tier microservices web application using Node.js, JWT, and a normalized database, allowing efficient project tracking.

Assess Buddy | [Github](#)

Coded a CLI tool using Bash to streamline the extraction, compilation, and execution of source code inside compressed folders, automating the evaluation of student-submitted assignments.